

AI-Powered Real Estate Intelligence, Valuation & Investment Recommendation Platform

Feature Engineering Report

Type	Regression + Recommendation System — Real Estate Intelligence Platform
Dataset	Indian Real Estate Housing Dataset (Kaggle) + Census Population Density Data
Dataset URL	kaggle.com/datasets/ankushpanday1/india-house-price-prediction
Records	2,50,000 Property Records
Features	34 Features (23 Original + 11 Engineered)
Best Model	XGBoost Regressor
Test Accuracy	R ² Score = 0.9769
MAE RMSE	13.68 Lakhs 18.03 Lakhs
Deployment	Streamlit web application
Notebooks	02_feature_engineering

This Feature Engineering Report describes the transformation and creation of meaningful features to enhance the predictive performance of the project's machine learning models. The process included encoding categorical variables, scaling numerical features, generating domain-specific indicators, and creating new features related to location, infrastructure, and property characteristics. These engineered features improved model accuracy, interpretability, and the overall effectiveness of the real estate intelligence platform.

1. Introduction

Feature Engineering is the process of using domain knowledge to extract new variables from raw data, transforming variables to better represent the underlying problem, and encoding categorical data for machine learning algorithms. This phase started with a cleaned dataset of 250,000 records across 20 columns and concluded with a highly optimized matrix ready for predictive modeling.

2. New Features Created

Domain-specific features were engineered to provide the ML models with deeper contextual data:

- **Bathrooms & Balcony:** Derived directly and probabilistically from the BHK count.
- **Population_Density:** Programmatically mapped from Census 2011 data based on the City and State columns. Mean density: 4737.35 people/sq.km.
- **Mall_Distance_km & Crime_Index:** Simulated based on city tier logic and density to reflect urban realities. Mean mall distance is 7.9km; mean crime index is 4.26.
- **Infra_Growth_Score:** A composite score derived from Nearby_Schools, Nearby_Hospitals, and Public_Transport_Accessibility. Mean score: 5.57.
- **Amenity_Count:** Extracted by parsing and counting the comma-separated strings in the original 'Amenities' column.
- **Floor_Ratio:** Calculated as $\text{Floor_No} / \text{Total_Floors}$, representing the relative height of the apartment in the building. Mean ratio: 0.49.
- **Price_Category:** A binning of prices into Budget, Mid, Premium, and Luxury brackets for classification purposes.

3. Encoding

Machine learning models require numerical input. The following encoding strategies were applied:

Label/Target Encoding:

- Applied to high-cardinality categorical variables: `State` and `City`. These were encoded based on the mean target price of the region to preserve the geographical pricing hierarchy without creating hundreds of sparse columns.

One-Hot Encoding (OHE):

- Applied to nominal categorical columns: `Property_Type`, `Furnished_Status`, `Facing`, and `Owner_Type`. This expanded these features into boolean (0/1) columns (e.g., `Property_Type_Apartment`, `Property_Type_Villa`).

After encoding and dropping redundant string columns, the dataset expanded to 38 columns.

4. Scaling

To ensure algorithms (especially gradient descent-based ones) converge quickly and do not disproportionately weight large numerical values (like `Size_in_SqFt`) over smaller ones (like `BHK`), feature scaling was applied.

- **Min-Max Scaling:** Scaled all features to a strict 0 to 1 range. This preserves the exact shape of the original distribution and is highly effective for Tree-based algorithms.
- **Standard Scaling:** Centered data around a mean of 0 with a standard deviation of 1.

Decision: The Min-Max scaled version was saved as the primary dataset for modeling, as tree-based models (XGBoost, Random Forest) handle non-normal distributions well but benefit from bounded inputs.

5. Feature Selection

To optimize model training time and reduce noise, feature selection was performed based on correlation and predictive power.

Correlation-Based Selection:

Features with very low predictive power or high multicollinearity were dropped. The original string-based 'Amenities' column was dropped in favor of the numerical 'Amenity_Count'.

Top Features Selected (By Target Correlation):

Feature	Correlation with Price
Size_in_SqFt	0.744
Population_Density	0.460
Crime_Index	0.334
Mall_Distance_km	0.290 (Absolute)
Availability_Status	0.214
BHK	0.103
Bathrooms	0.103
Infra_Growth_Score	0.102

6. Final Dataset Summary

Shape after Feature Engineering: 250,000 rows × 35 columns.

Final Features List (34 Predictors + 1 Target):

BHK, Size_in_SqFt, Floor_No, Total_Floors, Age_of_Property, Nearby_Schools, Nearby_Hospitals, Public_Transport_Accessibility, Parking_Space, Security, Availability_Status, Bathrooms, Balcony, Population_Density, Mall_Distance_km, Crime_Index, Infra_Growth_Score, Amenity_Count, Floor_Ratio, State_Encoded, City_Encoded, Property_Type_Apartment, Property_Type_Independent House, Property_Type_Villa, Furnished_Status_Furnished, Furnished_Status_Semi-furnished, Furnished_Status_Unfurnished, Facing_East, Facing_North, Facing_South, Facing_West, Owner_Type_Broker, Owner_Type_Builder, Owner_Type_Owner.

Target Variable: Estimated_Price

7. Conclusion

The Feature Engineering pipeline successfully transformed raw, unstructured data (like comma-separated amenities and textual city names) into a rich, numerical matrix. With 34 highly correlated and scaled predictive features, the dataset is now fully optimized and ready for Phase 6: Machine Learning Model Development.